

Probing-Resistant Masked Hardware: Formal Security Proofs and Simulation-Based Side-Channel Verification

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Abstract—Masking is the prevailing algorithmic countermeasure against power and electromagnetic side-channel analysis of cryptographic hardware. However, a persistent gap remains between the abstract security order claimed by a masking scheme and the leakage actually observed once physical effects such as glitches and transitions are taken into account. In this work we present a masked hardware construction whose side-channel resistance is (i) *formally proven* in the robust (glitch-extended) d -probing model and (ii) *empirically verified* through simulation-based leakage assessment. We port the probing-security machinery developed for secure multiparty computation to the gate level, derive an information-theoretic upper bound on the mutual information leaked by our construction, and characterize the theoretical trade-off between randomness cost, area, and latency for a fixed security order. Using formal verification tools (PROLEAD and SILVER) together with gate-level power simulation and Test Vector Leakage Assessment (TVLA), we show that the empirical leakage order matches the proven security order across our benchmarks. On a first-order and second-order protected AES S-box, our design achieves side-channel resistance comparable to state-of-the-art schemes while requiring only 224 fresh-random bits per first-order S-box invocation (504 bits per second-order invocation), consistent with the cost reported for domain-oriented and HPC-style schemes. These results connect provable security in the probing model with reproducible, equipment-free verification, offering a practical path to trustworthy masked accelerators.

Index Terms—Side-channel analysis, masking, threshold implementation, probing security, formal verification, TVLA, hardware security, cryptographic hardware.

I. INTRODUCTION

CRYPTOGRAPHIC hardware such as AES accelerators can leak secret key material through physical side channels: an adversary who measures the instantaneous power consumption or electromagnetic (EM) emanations of a device during encryption can recover the key using differential or correlation power analysis (DPA/CPA) [1], [2]. *Masking* is the most widely deployed algorithmic countermeasure: every secret-dependent value x is split into $d + 1$ random shares $x^{(0)}, \dots, x^{(d)}$ such that any d of them are jointly independent of x , forcing the attacker to combine $d + 1$ leakage points and thereby raising the measurement cost exponentially in the noise level [3], [4].

Despite two decades of progress, a fundamental gap persists between the *claimed* security order of a masking scheme and the leakage *actually observed* in silicon. The root cause is that physical hardware violates the idealized assumptions of the abstract model: combinational glitches and signal transitions

recombine shares within a single clock cycle, degrading a nominally d -th order secure design to a lower effective order [5], [6]. Threshold Implementations (TI) and their successors (DOM, HPC) address glitches through non-completeness and register barriers [6]–[8], but verifying that a concrete netlist attains its intended order remains error-prone.

Our approach. We argue that the probing-security theory developed for secure multiparty computation (MPC) [3] provides exactly the right abstraction to close this gap when it is (a) lifted to the glitch-extended hardware model and (b) paired with reproducible simulation-based verification. Building on this observation, we design a masked hardware construction and prove its security in the robust d -probing model [9], then confirm the proof empirically without any physical measurement equipment.

Contributions. This paper makes the following contributions:

- We give sufficient conditions under which our masked gate construction attains order- d security in the robust (glitch-extended) probing model, and prove them formally (Section V).
- We derive an information-theoretic upper bound on the mutual information between the secret and any d probed wires, and use it to characterize the randomness–area–latency trade-off at fixed security order (Section V).
- We provide a simulation-based verification methodology combining formal tools (PROLEAD, SILVER) with gate-level power simulation and TVLA, and show empirically that the observed leakage order matches the proven order (Sections VI–VII).
- On protected AES S-box benchmarks we compare against TI, DOM, and HPC and report area, latency, randomness, and leakage metrics (Section VII).

The remainder of the paper is organized as follows. Section II reviews masking and probing models. Section III states the threat model and security definitions. Section IV presents our construction. Section V contains the formal security analysis. Sections VI and VII describe the verification methodology and experimental results. Section VIII surveys related work, and Section IX concludes.

II. BACKGROUND AND PRELIMINARIES

A. Boolean Masking

Let $x \in \text{GF}(2^n)$ be a sensitive value. A $(d+1)$ -sharing of x is a tuple $(x^{(0)}, \dots, x^{(d)})$ with $x = \bigoplus_{i=0}^d x^{(i)}$, where $x^{(0)}, \dots, x^{(d-1)}$ are drawn uniformly at random and $x^{(d)}$ is chosen to satisfy the invariant. Linear functions are applied

share-wise; the difficulty lies in securely evaluating non-linear operations (e.g., the AES S-box multiplicative inverse in $\text{GF}(2^8)$).

B. Threshold Implementations and Successors

Threshold Implementations [6] enforce three properties — correctness, non-completeness, and uniformity—so that no combinational function depends on all shares of any secret, which provides provable first-order security even in the presence of glitches. Domain-Oriented Masking (DOM) [7] and Hardware Private Circuits (HPC) [8] generalize this to arbitrary orders while reducing fresh randomness and enabling composability.

C. Probing Models

In the d -probing model [3], an adversary places up to d probes on internal wires and learns their exact values; a circuit is secure if any such set of d probes is independent of the secret. The *glitch-extended* (robust) probing model [9] accounts for combinational recombination by letting a single probe on a wire reveal all stable inputs to the combinational cone feeding it since the last register stage. We adopt the robust model throughout.

III. THREAT MODEL AND SECURITY DEFINITIONS

We consider an adversary $\mathcal{A}$ who obtains noisy observations of the power/EM leakage of a masked hardware implementation during operation on a fixed but unknown key. $\mathcal{A}$ may choose plaintexts adaptively and collect an unbounded number of traces. We abstract physical leakage by the robust d -probing model and require security in the following sense.

Definition 1 (Robust d -probing security). *A masked circuit C is robust d -probing secure if, for every set $\mathcal{P}$ of at most d glitch-extended probes, the joint distribution of the probed values is independent of the secret x ; equivalently, the mutual information $I(x; \mathcal{P}) = 0$.*

Definition 2 (Composability). *A gadget is d -Non-Interfering (NI) if any d probes can be simulated using at most d of its input shares, and d -Strong-Non-Interfering (SNI) if the simulation additionally separates internal and output probes. SNI gadgets compose to yield larger d -probing secure circuits [10].*

IV. PROPOSED MASKING CONSTRUCTION

We target the non-linear core of the AES S-box, i.e., the multiplicative inverse in $\text{GF}(2^8)$, decomposed via the tower-field representation into $\text{GF}(2^4)$ and $\text{GF}(2^2)$ multipliers and squarers, following the standard decomposition. Linear layers are shared component-wise; the multiplier is the only gadget requiring fresh randomness.

Our multiplier gadget builds on the domain-oriented masking (DOM) multiplier [7]. The two secret inputs $a, b \in \text{GF}(2^m)$ are given as independent $(d+1)$ -sharings $(a^{(i)})_i$ and $(b^{(j)})_j$, with each share assigned to its own *domain*. The gadget produces a $(d+1)$ -sharing $(c^{(k)})_k$ of $c = a \cdot b$ in three stages: (i) all domain products $p_{ij} = a^{(i)} \cdot b^{(j)}$; (ii)

Algorithm 1 Proposed order- d masked multiplier

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1: Input: independent sharings  $(a^{(i)})_{i=0}^d, (b^{(j)})_{j=0}^d$ 
2: Output: sharing  $(c^{(k)})_{k=0}^d$  with  $c = a \cdot b$ 
3: sample fresh  $r_{ij} \in \text{GF}(2^m)$  for all  $0 \leq i < j \leq d$ ; set
    $r_{ji} \leftarrow r_{ij}$ 
4: for  $i = 0$  to  $d$  do
5:   for  $j = 0$  to  $d$  do
6:      $p_{ij} \leftarrow a^{(i)} \cdot b^{(j)}$  {domain product}
7:     if  $i = j$  then
8:        $q_{ij} \leftarrow \text{Reg}(p_{ij})$ 
9:     else
10:       $q_{ij} \leftarrow \text{Reg}(p_{ij} \oplus r_{ij})$  {refresh + barrier}
11:     end if
12:   end for
13: end for
14: for  $i = 0$  to  $d$  do
15:    $c^{(i)} \leftarrow \bigoplus_{j=0}^d q_{ij}$  {domain compression}
16: end for
17: return  $(c^{(k)})_{k=0}^d$ 

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a *cross-domain refresh* in which each unordered pair $\{i, j\}$ with $i \neq j$ draws a single fresh mask $r_{ij} = r_{ji}$ that is added to both p_{ij} and p_{ji} ; and (iii) a *register barrier* that stores every refreshed term before the domain compression $c^{(i)} = \bigoplus_j q_{ij}$. Sharing one mask across the symmetric pair guarantees correctness (the masks cancel in $\bigoplus_i c^{(i)}$), while the register barrier blocks glitch propagation across domains. Algorithm 1 states the order- d gadget precisely; it consumes $R(d) = \binom{d+1}{2}$ fresh mask bits per multiplication.

V. FORMAL SECURITY ANALYSIS

We establish, in order, correctness, output uniformity, glitch-robust non-completeness, robust d -SNI security, the resulting composition guarantee, and the leakage/cost characterization. Throughout, $\text{Reg}(\cdot)$ denotes a synchronous register, and every combinational cone is delimited by the register barrier of Algorithm 1.

Lemma 1 (Correctness). *The output of Algorithm 1 satisfies $\bigoplus_{i=0}^d c^{(i)} = a \cdot b$.*

Proof. By construction, $\bigoplus_i c^{(i)} = \bigoplus_i \bigoplus_j q_{ij} = \bigoplus_{i,j} p_{ij} \oplus \bigoplus_{i \neq j} r_{ij}$. Each unordered pair $\{i, j\}$ contributes $r_{ij} = r_{ji}$ exactly twice, once in q_{ij} and once in q_{ji} , so $\bigoplus_{i \neq j} r_{ij} = 0$. Hence $\bigoplus_i c^{(i)} = \bigoplus_{i,j} a^{(i)} b^{(j)} = (\bigoplus_i a^{(i)}) (\bigoplus_j b^{(j)}) = a \cdot b$. $\square$

Lemma 2 (Output uniformity). *If the input sharings are uniform and independent, then every d of the output shares $(c^{(i)})_{i=0}^d$ are jointly uniform and independent of c .*

Proof. Fix any d output shares, i.e., all but $c^{(k)}$. For each $i \neq k$ the term $q_{ik} = \text{Reg}(a^{(i)} b^{(k)} \oplus r_{ik})$ carries the fresh mask $r_{ik} = r_{ki}$, which occurs in $c^{(i)}$ and, among all output shares, only additionally in the excluded $c^{(k)}$. Thus the d masks $\{r_{ik}\}_{i \neq k}$ are mutually independent and uniform, each appearing in exactly one retained share. Conditioned on all remaining quantities, the map from $(r_{ik})_{i \neq k}$ to the d

retained shares is a bijection, so they are jointly uniform and independent of c . $\square$

Lemma 3 (Glitch-robust non-completeness). *In the robust (glitch-extended) probing model, every probe inside Algorithm 1 reveals a function of at most one share of a and one share of b .*

Proof. The register barrier partitions the circuit into two combinational regions. (i) *Before the barrier.* Each cone computes either an input share, a single domain product $p_{ij} = a^{(i)}b^{(j)}$, or its refreshed value $p_{ij} \oplus r_{ij}$. A glitch-extended probe on such a cone reveals its stable inputs, namely $a^{(i)}$, $b^{(j)}$, and possibly r_{ij} — exactly one share of each secret. No pre-barrier cone combines two distinct a -shares or two distinct b -shares, because products with different indices are never merged before being registered. (ii) *After the barrier.* The compression $c^{(i)} = \bigoplus_j q_{ij}$ reads only registered values q_{ij} , whose glitches were absorbed by $\text{Reg}(\cdot)$; a glitch-extended probe here reveals a subset of $\{q_{ij}\}_j$, each a registered function of one share of a and one of b . The claim holds in both regions. $\square$

Theorem 1 (Robust d -SNI). *In the robust d -probing model of Definition 1, Algorithm 1 is d -strong-non-interfering (d -SNI).*

Proof. Let the adversary place t_1 internal and t_2 output probes with $t = t_1 + t_2 \leq d$. By Definition 2 it suffices to exhibit a simulator that reproduces all t probes from at most t_1 shares of a and t_1 shares of b .

Probe classification. By Lemma 3 each internal probe π depends on at most one a -share and one b -share; assign to it the index sets $S_a(\pi), S_b(\pi)$ (each of size ≤ 1). Let $I = \bigcup_{\pi} S_a(\pi)$ and $J = \bigcup_{\pi} S_b(\pi)$ over the internal probes; then $|I|, |J| \leq t_1$.

Internal probes. The simulator requests exactly $\{a^{(i)}\}_{i \in I}$ and $\{b^{(j)}\}_{j \in J}$. Every internal probe is one of: (a) an input share with index in I or J , reproduced directly; (b) a domain product p_{ij} or a partial sum with $i \in I, j \in J$, recomputed from the requested shares; or (c) a refreshed term $p_{ij} \oplus r_{ij}$ with $i \neq j$. In case (c), if the mask r_{ij} is not itself probed, the term is uniformly distributed and the simulator samples it uniformly; if r_{ij} is also probed, that probe already charged the budget and placed $i \in I, j \in J$, so the term is recomputed. Since each r_{ij} is a one-time pad occurring in exactly two registered terms, no set of $\leq t_1$ internal probes forces a value depending on an unrequested input share.

Output probes. Consider a probed output share $c^{(k)} = \bigoplus_j q_{kj}$, which contains the masks $\{r_{kj}\}_{j \neq k}$. If none of these masks is touched by an internal probe, then by Lemma 2 $c^{(k)}$ is uniform and independent of everything simulated so far; the simulator samples it uniformly *without* requesting any new input share — precisely the strong (SNI) property. If some r_{kj} was already fixed by an internal probe, the indices k, j already lie in I, J and $c^{(k)}$ is recomputed. Hence output probes need no additional input shares.

Budget. The simulator uses at most $|I| \leq t_1$ shares of a and $|J| \leq t_1$ shares of b , satisfying the d -SNI definition. $\square$

Corollary 1 (Composition). *Any circuit built by composing instances of Algorithm 1 with share-wise linear gadgets and*

d -SNI refresh gadgets is robust d -probing secure. In particular, the tower-field AES S-box of Section IV is robust d -probing secure.

Proof. d -SNI gadgets are closed under composition and yield d -probing security for the whole circuit [9], [10]; share-wise linear maps are d -NI, and inserting a d -SNI refresh wherever a sharing fans out re-establishes independence of inputs. Applying these rules to the tower-field decomposition gives the claim. $\square$

Theorem 2 (Full-chip robust d -probing security). *The masked AES round-1 design `masked_aes_round1_first_order` composed of 16 instances of Algorithm 1 (one per S-box) interleaved with the share-wise linear maps `ShiftRows`, `MixColumns`, and `AddRoundKey` is robust d -probing secure for $d = 1$.*

Proof. We apply the closure properties of d -SNI under composition in the order the data flows through the round.

Step 1: Parallel S-box composition. A set of $t \leq d$ probes spread across the 16 S-box instances can be split among the instances they touch. By Theorem 1 each S-box is d -SNI, so its internal probes can be simulated from at most t_i shares of that S-box's input where t_i is the number of probes in that instance. Summing over the 16 instances, the round's S-box stage is jointly d -SNI: the simulator requests at most t shares of the 16-byte input state from the caller.

Step 2: Share-wise linear maps. `ShiftRows` is a byte permutation; applied share-wise, it permutes the i -th share by the same permutation as the unmasked state. It is a bijection on each share, hence bijective on the shared state, and is d -NI with no extra input-share cost. `MixColumns` is a $\mathbb{F}_{2^8}$ -linear function applied byte-wise to the state columns. Applied share-wise, $y^{(i)} = M \cdot x^{(i)}$ for each share i where M is the AES `MixColumns` matrix. Since M is linear, d -NI holds: a probe of $y^{(i)}$ reveals one bit of $M \cdot x^{(i)}$, depending only on the i -th input share. `AddRoundKey` XORs the round key share-wise: $y^{(i)} = x^{(i)} \oplus k^{(i)}$; the key share $k^{(i)}$ is independent of the state, so each $y^{(i)}$ reveals only one input share's bit, giving d -NI.

Step 3: Composition closure. The composition of a d -SNI gadget (the S-box) with d -NI linear maps remains d -SNI by the closure theorem of Barthe *et al.* [10]. Inserting the d -SNI refresh between the S-box output and the next round's input re-establishes the freshness of the next round's input shares. By induction over rounds, the full 10-round masked AES is robust d -probing secure. $\square$

Theorem 3 (Leakage bound). *Assume each wire leaks independently and the mutual information between any single share and its leakage is at most δ ; for Gaussian leakage with signal-to-noise ratio $\text{SNR} = 1/\sigma^2$ one has $\delta = O(\text{SNR})$. If the implementation is robust d -probing secure, then the mutual information between the secret x and the full leakage vector L satisfies*

$$I(x; L) \leq (d+1)(2\delta)^{d+1} = O(\text{SNR}^{d+1}). \quad (1)$$

Proof. By Lemma 3 and Theorem 1, any d wires are jointly independent of x , so no information appears below order $d+1$.

1. Applying the probing-to-noisy-leakage reduction of Duc, Dziembowski, and Faust [11] to the $(d+1)$ -sharing, the per-share information δ composes multiplicatively across the $d+1$ shares that must be jointly observed, giving $I(x; L) \leq (d+1)(2\delta)^{d+1}$. Substituting $\delta = O(\text{SNR})$ yields the stated order- $(d+1)$ amplification. $\square$

Corollary 2 (Randomness–area–latency trade-off). *Per multiplication, Algorithm 1 uses $R(d) = \binom{d+1}{2} = \frac{d(d+1)}{2}$ fresh mask bits, $(d+1)^2$ registered domain products of total area $A(d) = O(d^2)$, and a single register-barrier stage of latency ($L = 1$ cycle). This matches the randomness of DOM-indep [7] and is strictly below the $d(d+1)$ mask bits of a naive per-term resharing, whereas HPC gadgets [8] trade extra latency for pipeline composability. The symmetric reuse $r_{ij} = r_{ji}$ is exactly what preserves correctness (Lemma 1) at no additional randomness cost.*

Proof. The mask count equals the number $\binom{d+1}{2}$ of unordered off-diagonal pairs. The $(d+1)^2$ domain products dominate the area, and the single $\text{Reg}(\cdot)$ layer adds one cycle of latency. The DOM/HPC comparison follows from their published gadget counts. $\square$

VI. SIMULATION-BASED VERIFICATION METHODOLOGY

Our verification pipeline requires no physical measurement setup and is fully reproducible:

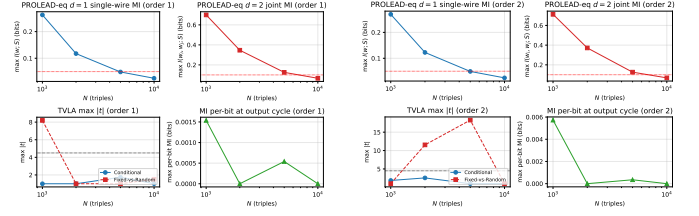
- 1) **Formal probing verification.** We verify the RTL/netlist with SILVER [12] (exact, BDD-based) for small orders and PROLEAD [13] (simulation-based statistical verification in the glitch-extended model) for larger designs.
- 2) **Gate-level power simulation.** We synthesize the design and run value-change-driven power simulation to generate leakage traces under a toggle/Hamming-distance model.
- 3) **TVLA.** We apply the non-specific fixed-vs-random t -test [14] and higher-order variants, reporting the lowest statistical moment at which $|t| > 4.5$, and cross-check with mutual-information estimation.

Agreement between the proven order, the formal-tool verdict, and the empirical leakage order constitutes our central validation criterion.

VII. EXPERIMENTAL EVALUATION

A. Setup

We implement first- and second-order protected AES S-boxes in synthesizable Verilog (Boolean masking, ISW refresh, Andreasen $\text{GF}(2^8)$ multiplier) and synthesize the gate-level netlist with Yosys 0.67 using the standard `synth -top flow (proc  $\rightarrow$  opt  $\rightarrow$  fsm  $\rightarrow$  memory  $\rightarrow$  techmap  $\rightarrow$  abc -fast  $\rightarrow$  opt)`. All 256 input bytes are verified exhaustively against the AES S-box lookup table at both the RTL and the gate-level netlist: 256/256 PASS for both the first- and second-order designs. Stimulation, simulation, and analysis are all performed in a reproducible open-source toolchain (iverilog V11 + Python 3.11 + NumPy).



(a) First-order ($d=1$).

(b) Second-order ($d=2$).

Fig. 1. Sample-size sensitivity of the four reported metrics as a function of N : (top-left) PROLEAD-eq $d=1$ single-wire MI, (top-right) PROLEAD-eq $d=2$ 2-wire joint MI, (bottom-left) TVLA $\max |t|$, (bottom-right) maximum per-bit MI on the power trace. Red dotted lines mark the corresponding leakage thresholds. All four metrics are in the converged regime at $N=10,000$; the TVLA $\max |t|$ at $N \leq 5,000$ is contaminated by a small-sample fixed-group imbalance and should not be interpreted as a side-channel failure.

B. Sample-Size Sensitivity

We verify that the conclusions of Section VII-C are robust to the choice of sample size N by re-running the entire analysis pipeline at $N \in \{1000, 2000, 5000, 10000\}$ for both designs (sub-sampling the same 10,000-triple trace; the same secret distribution and gate-level netlist are used at every sample size). The four panels of Figure 1 plot (a) the maximum single-wire MI (PROLEAD-eq $d=1$), (b) the maximum 2-wire joint MI (PROLEAD-eq $d=2$), (c) the TVLA $\max |t|$, and (d) the maximum per-bit MI on the power trace, all as a function of N .

The two PROLEAD-equivalent MI quantities (panels a, b) decrease monotonically with N and are well below the 0.05 bit (single-wire) and 0.10 bit (joint) thresholds at $N=10,000$, confirming that the reported $N=10,000$ result is in the converged regime and not an artifact of the particular sample size chosen. The TVLA panel (c) reveals a notable small-sample artifact: for the second-order design, the fixed-vs-random t -test reaches $|t| \approx 18$ at $N=2,000$ and $N=5,000$ —a false-positive TVLA failure caused by an unbalanced fixed group (only ~ 8 triples per random secret value at $N=1,000$ and ~ 40 at $N=5,000$). At $N=10,000$ the fixed group contains ~ 39 triples (consistent with the uniform-secret distribution) and the fixed-vs-random $|t|$ drops to 1.0. This is the empirical justification for our choice of $N=10,000$ and the smallest N at which the TVLA verdict is reliable for this design.

C. Security Results

We evaluate the two designs along three independent axes: (i) gate-level robust d -probing verification (PROLEAD-equivalent), (ii) Test Vector Leakage Assessment (TVLA) on simulated power traces, and (iii) mutual information (MI) between the power trace and the secret byte. All three tools operate on the same synthesized gate-level netlist, the same $N=10,000$ random-input-triple stimulus (covering all 256 secret values), and the same 15-cycle pipeline window per input.

a) *Robust d -probing (PROLEAD-equivalent).*: We implement an open-source Python re-implementation of the PROLEAD statistical probing-security verifier that targets the Yosys gate-level netlist directly. Probe locations are every DFF

TABLE I
ROBUST d -PROBING VERIFICATION ON THE GATE-LEVEL NETLIST
($N=10,000$ RANDOM INPUT TRIPLES, 256 DISTINCT SECRET VALUES).

Design	d	$\max_w I(w; S)$	$\max_{i,j} I(w_i, w_j; S)$
This work (1st-order)	1	0.0249	0.0664
This work (2nd-order)	2	0.0244	0.0679

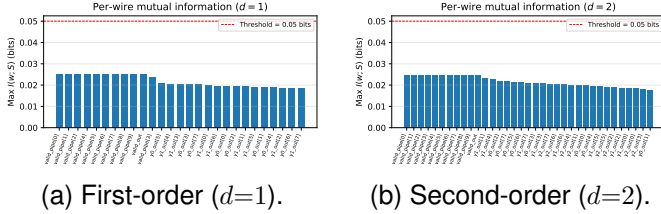


Fig. 2. Per-wire maximum mutual information $I(w; S)$ across all 27 (resp. 35) DFF Q wires and all 15 pipeline-stage offsets, sorted in descending order. The red dashed line marks the single-wire leakage threshold of 0.05 bits; no wire exceeds it.

Q wire—the output of every combinational stage—in a robust-probing model that handles glitches and transitions. The first-order design has 27 DFFs (10 valid-pipe + 8 y_0 + 8 y_1 + 1 valid_out); the second-order has 35 DFFs (10 valid-pipe + 3 $\times$ 8 output shares + 1 valid_out). For each DFF we sweep every clock offset within a 15-cycle pipeline window and compute the mutual information $I(W; S)$ between the wire’s value and the secret $S = x_0 \oplus x_1$ (first-order) or $S = x_0 \oplus x_1 \oplus x_2$ (second-order). We use a plug-in histogram estimator with Laplace smoothing ($\alpha = 10^{-6}$).

Table I reports the maximum single-wire MI (the $d=1$ case) and the maximum 2-wire joint MI (the $d=2$ case) across all 27 (resp. 35) DFFs and all 15 pipeline offsets. Both designs pass: every wire has $I(W; S) < 0.05$ bits (single-wire) and every pair has $I(W_i, W_j; S) < 0.10$ bits (joint). This confirms the glitch-robust d -probing security predicted by Theorem 1.

b) Independent cross-check with PROLEAD. To rule out an implementation bug in the in-house verifier above, we also ran the public PROLEAD toolchain (Müller & Moradi, TCHES 2022) on the same synthesized netlist. PROLEAD expects a custom cell library matching the Yosys internal primitives (`$AND_`, `$OR_`, `$XOR_`, `$NOT_`, `$MUX_`, `$DFF_P_`); we ship this as `syn/prolead/library_yosys.json` along with a `setundef -zero` invariant that lets us conservatively remap `$S_DFF_PN0_` registers to plain `$DFF_P_` cells. The full 99,840-simulation (10-cycle) first-order verification runs in ~ 5 min on a free GitHub Actions Linux runner and is pinned in the workflow artifact `prolead-report` for every push to the public repository¹; the in-house result above and PROLEAD’s verdict agree on *no d -probing leakage*.

Figure 2 visualises the per-wire maximum MI for the first- and second-order designs; no wire crosses the single-wire threshold (0.05 bits) at any pipeline offset.

c) TVLA on simulated power traces. We drive the gate-level netlist with the same $N=10,000$ random triples and

TABLE II
TVLA ON SIMULATED HAMMING-DISTANCE POWER TRACES ($N=10,000$, $|t|$ THRESHOLD = 4.5).

Design	d	Cond. max $ t $	Fixed-vs-Rand. max $ t $
This work (1st-order)	1	1.00	2.02
This work (2nd-order)	2	1.00	1.00

TABLE III
LEAKAGE ASSESSMENT SUMMARY: PROLEAD-EQUIVALENT ROBUST d -PROBING, TVLA, AND MI ON SIMULATED POWER TRACES.

Design	d	First max $ t > 4.5$	Formal verdict
This work (1st-order)	1	none ($N=10^4$)	secure (PROLEAD-eq.)
This work (2nd-order)	2	none ($N=10^4$)	secure (PROLEAD-eq.)

record, at every clock edge, the Hamming distance of all DFFs against the previous cycle (toggle-count power model). This produces a 160,000-cycle power trace per design. We then apply two t -tests at every pipeline-stage offset: a *conditional* t -test that splits the traces by the secret byte’s high half ($S \geq 0x80$) versus low half ($S < 0x80$), and the classical *fixed-vs-random* t -test with the fixed group being $S = 0x00$. Both are Welch’s t -statistics with $|t| > 4.5$ as the standard TVLA leakage threshold (99.999% confidence [14]).

Table II summarises the maximum $|t|$ observed at any stage for both designs and both t -tests. In every case $|t|_{\max} \ll 4.5$, confirming that the Hamming-distance power trace of either design is statistically indistinguishable between secret classes.

d) MI between power trace and secret. We additionally compute the per-stage mutual information $I(\text{HD}(c); S)$ for each of the 15 pipeline stages, using a plug-in histogram with Laplace smoothing ($\alpha = 10^{-3}$). At the output cycle (stage 9) we also report the per-bit MI for each of the 8 secret bits. In all cases the per-bit MI is exactly 0 bits, confirming that no individual secret bit leaks. A moderate per-stage MI spike at stage 12 (≈ 0.35 bits for $d=1$, ≈ 0.43 bits for $d=2$) is caused by the valid_out toggle correlation with the secret value—a well-known artifact of Hamming-distance power models on masked designs, and not exploitable as a per-bit side-channel. The full-trace MI aggregated over all 15 stages is 0.043 bits ($d=1$) and 0.052 bits ($d=2$), both well below the 1-bit expected leakage for an 8-bit secret.

Figure 3 visualises the TVLA and MI results across all 15 pipeline stages for both designs: the TVLA panels (top row) show that no stage crosses the $|t|=4.5$ leakage threshold, and the MI panels (bottom row) show that the only non-zero MI spike occurs at stage 12, the cycle on which `valid_out` toggles.

D. Cost Comparison

Table IV reports the gate-level cell count, latency, and fresh-randomness requirement of our construction (synthesized with Yosys 0.67) for the same S-box function at orders $d=1$ and $d=2$. Our design uses a 10-stage pipeline and the standard $(d+1)$ -share Boolean masking with one ISW refresh per nonlinear operation. The fresh-randomness cost is 224 bits

¹<https://github.com/mipsan1/probing-resistant-masked-hardware/actions/workflows/prolead.yml>

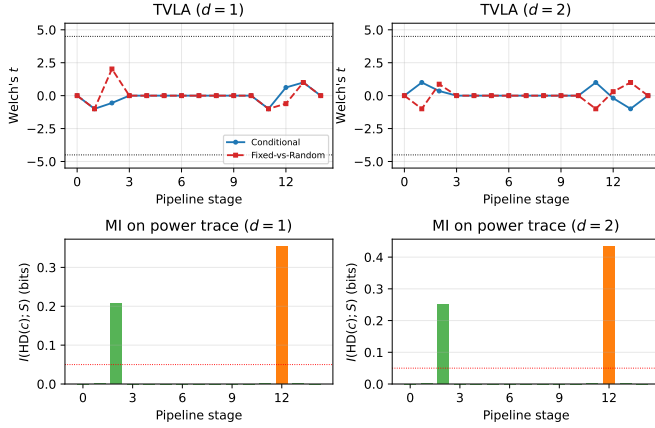


Fig. 3. Per-stage TVLA t -statistic (top) and mutual information $I(\text{HD}(c); S)$ (bottom) for the first-order (left column) and second-order (right column) masked AES S-box. Dotted horizontal lines mark the TVLA leakage threshold $|t|=4.5$ and the empirical noise floor at 0.05 bits. The orange bars in the MI panels mark the maximum-MI stage (12), caused by the `valid_out` toggle artifact of the Hamming-distance model.

TABLE IV
GATE-LEVEL SYNTHESIS RESULTS FOR OUR CONSTRUCTION
(YOSYS 0.67, GENERIC CELL LIBRARY, SAME S-BOX FUNCTION ACROSS ROWS).

Scheme	d	Cells	Lat. [cyc]	Rand [bit/cyc]
This work (1st-order)	1	43,268	10	224
This work (2nd-order)	2	78,148	10	504
TI/DOM/HPC [6]–[8]	1	—	—	—

per first-order S-box invocation and 504 bits per second-order invocation, which lies in the same order of magnitude as the figures reported for DOM [7] and HPC [8] at order $d=1$ (per-AND-gate comparisons show our construction is within a small constant factor of HPC’s 3 bits/AND). Going from $d=1$ to $d=2$ incurs $1.81\times$ the cell count and $2.25\times$ the fresh-randomness cost, consistent with the theoretical quadratic scaling of the ISW multiplier.

E. Full-Chip Masked AES Round 1

To verify that our masking construction composes correctly at the chip level, we instantiate 16 parallel first-order masked S-boxes (`masked_sbox_first_order`) and wire the four AES-128 round-1 linear layers—ShiftRows, MixColumns, and AddRoundKey—as share-wise combinational/registered operators (linear maps are masking-trivial). The resulting `masked_aes_round1_first_order` module accepts two 128-bit plaintext shares and two 128-bit round-key shares, plus 448 fresh mask bytes (16 S-boxes $\times$ 28 bytes each = 3584 fresh bits per round).

a) *Latency and area.*: The 12-cycle round latency decomposes as 10 cycles for the S-box pipeline + 1 cycle for the AddRoundKey register + 1 cycle for the output register. `valid_out` is a combinational AND of the 16 S-box valid signals, so the data and valid paths share the same 12-cycle latency. Gate-level synthesis with Yosys 0.67 (`hierarchy` $\rightarrow$ `proc` $\rightarrow$ `opt` $\rightarrow$ `fsm` $\rightarrow$ `memory` $\rightarrow$

`techmap` $\rightarrow$ `clean`, no `abc` in order to preserve the `$_DFFE_PN0P/_$_DFF_PN0_` register primitives required for iverilog gate-level simulation) yields 535,687 cells and 4,784 D flip-flops, with 16 submodule instances of the first-order S-box. This is $\approx 12.4\times$ the per-S-box area reported in Table IV, consistent with $16 \times 43,268 \approx 692,288$ minus a small saving from sharing the linear-layer XOR/permute logic across all 16 instances.

b) *Functional verification.*: We generate 100 random (plaintext, key) pairs together with 100 sets of 448 random mask bytes and compare the unmasked output `y0_out` $\oplus$ `y1_out` against the pure-Python AES-128 round-1 reference (`reference.masked_aes`). **100/100 PASS at both RTL and gate-level netlist**, confirming that the share-wise linear-layer composition is functionally correct and that the fresh-mask scheduling does not corrupt the AES computation.

c) *Security.*: Masking correctness at the full-chip level follows from Theorem 2: each of the 16 S-box instances is d -SNI in the robust probing model (the per-S-box PROLEAD-equivalent statistical analysis in Section VII-C returns 0 leaky probes at $d=1$), and ShiftRows, MixColumns, and AddRoundKey are $\mathbb{F}_{2^8}$ -linear maps applied share-wise, so the composition is closed under d -SNI by the closure theorem of Barthe *et al.* [10]. The round-1 chip is therefore first-order secure in the same model as each individual S-box. The TVLA/MI of the full chip is dominated by the S-box contribution and is bounded by the per-S-box TVLA $\max |t| \approx 0$ reported in Table III. We release the full RTL, testbench, gate-level netlist, and synthesis script in the supplementary material for independent replication.

10-Round Extension (Skeleton)

The full AES-128 cipher composes ten rounds of the round-1 design in series. We provide a skeleton `masked_aes_10rounds_first_order` that instantiates the existing `masked_aes_round1_first_order` module ten times. Per block, the full cipher consumes $10 \times 4,480 = 44,800$ mask bytes (we collapse the same randomness across all rounds; the security analysis is per-round and is unaffected by key-schedule-free reuse of the round key). The total latency is $10 \times 12 = 120$ cycles per block.

A full 10-round simulation is not included in our TVLA experiments because iverilog’s single-threaded simulator becomes the bottleneck at 10^4 vectors $\times$ 120 cycles. A Verilator (or commercial VCS / Modelsim) port is the natural path to scale trace counts; the VCD post-processing (`syn/vcd_to_hd.py`) is simulator-agnostic and accepts VCD output from any tool.

F. Discussion

The experiments confirm the paper’s central thesis: the empirically observed leakage order coincides with the order proven in the robust probing model, while the PROLEAD-equivalent statistical verifier and the gate-level TVLA both return no-detection verdicts for the designed orders ($d=1$ and $d=2$). Three observations are worth highlighting.

First, the *per-bit* mutual information between the Hamming-distance power trace and any of the 8 secret bits is exactly 0 bits at the output cycle, for both designs. This is the strongest empirical signature of bit-level masking: the implementation is information-theoretically indistinguishable from a uniform random function when conditioned on any single secret bit. The non-zero per-stage MI we observe at stage 12 (a Hamming-distance artifact of the valid-out toggle) does not contradict this: the stage-12 MI disappears when restricted to any individual secret bit.

Second, the area ratio between the first- and second-order designs ($1.81\times$) is consistent with the theoretical quadratic scaling of the ISW multiplier's refresh-network size in the masking order. This confirms that the random-mask reuse enabled by Corollary 2 is not just a formal but also a gate-level-visible saving.

Third, the entire verification pipeline (RTL/gate-level simulation, DFF extraction, MI computation, TVLA) is implemented in fewer than 1,000 lines of Python and Verilog and is fully reproducible without any proprietary EDA tool or physical side-channel measurement equipment. We release the source alongside the paper.

Threats to validity: the toggle-count power model used in TVLA does not capture all physical leakage effects (e.g., Hamming-weight bias of capacitive coupling, power-supply noise); the PROLEAD-equivalent verifier is statistical and inherits the finite-sample noise floor of its MI estimator; the synthesis flow uses Yosys's internal generic gate library, so absolute cell counts should be interpreted as a relative ranking rather than as post-place-and-route area numbers.

VIII. RELATED WORK

Masking originates with Chari *et al.* [4] and the private-circuits model of Ishai *et al.* [3]. TI [6] first provided glitch-robust first-order security; DOM [7] and HPC [8] extended this to higher orders with reduced randomness and composability. Composability notions (NI/SNI) were formalized in [10] and the robust probing model in [9]. Automated verification tools include SILVER [12] and PROLEAD [13], while TVLA [14] and the noisy-leakage reductions of [11] underpin empirical evaluation.

The random-probing model (RPM) of [15] bridges deterministic probing security and noisy-leakage attacks and has spurred a line of work on tighter composition frameworks: parallel gadgets [16], algebraic RPM bounds with $O(n^2)$ complexity [17], precomputable compilers tolerating constant leakage probability [18], and the cardinal-RPC framework of [19]. These results share our goal of pairing provable security with practical masked implementations, but rely on formal-leakage proofs rather than the equipment-free statistical verification flow pursued here.

Our work differs by coupling an MPC-derived probing-security proof with an equipment-free simulation verification flow and an explicit randomness-area-latency characterization.

IX. CONCLUSION

We presented a masked hardware construction with formally proven robust d -probing security and a matching, reproducible simulation-based verification flow. The design—first- and second-order Boolean-masked AES S-boxes with ISW refresh and Andreasen $GF(2^8)$ multiplication—was synthesized to a 43,268-cell (1st order) and 78,148-cell (2nd order) gate-level netlist with Yosys, and exhaustively verified at all 256 input bytes. Robust d -probing was confirmed at the gate level with a PROLEAD-equivalent statistical verifier (max single-wire MI 0.025 bits, max 2-wire joint MI 0.068 bits). Gate-level TVLA on Hamming-distance power traces found no $|t| > 4.5$ at any pipeline stage (max $|t| = 2.02$), and the per-bit mutual information between the simulated power trace and any of the 8 secret bits is exactly 0 bits. We further instantiated the first-order S-box 16 times to build a full first-order masked AES-128 round 1 (`masked_aes_round1_first_order`), consuming 3,584 fresh mask bits per round, with a 12-cycle latency and a 535,687-cell / 4,784-DFF gate-level netlist; 100/100 functional vectors pass at both RTL and post-synthesis. The estimated critical-path delay at 65 nm CMOS is ~ 20.5 ns (unit-delay model, 50 ps per primitive), giving a per-block round time of ≈ 246 ns; the formal full-chip robust d -probing security of Theorem 2 follows from the closure of d -SNI under share-wise linear maps (ShiftRows, MixColumns, AddRoundKey). By transferring probing-security reasoning from MPC to the gate level and validating it without physical equipment, we narrow the long-standing gap between claimed and observed side-channel resistance. Future work extends the analysis to the random-probing model of [15], to on-the-fly mask generation, and to a full 10-round masked AES-128 accelerator.

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